

Stat 301 Notes

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Course Goals for STAT 301

STAT 301 is an introduction to sampling. By the end of the course, you will:

- Learn the basic methodology of how classical survey samples are conducted based on simple random sampling.
- Recognize sampling and non-sampling errors.
- Be able to use extra information in sampling via ratio estimators and regression estimators.
- Be able to define and use stratified samples and know why they are often better than SRS.
- Be able to design one and two stage cluster samples and know their key properties.
- Recognize strengths and weaknesses of different sampling approaches.
- Use standard R packages to analyze survey data and estimate population means, proportions, totals, and more.

1 Introduction